

PROIECTAREA CU MICROPROCESOARE

C3 2025-2026 Sem. 2

Proiect 2 Sapt 2 4 martie 2026 19:00-20:00

serban@upit.ro

Regulament disciplina

Nota finala este formata din activitatile:

- Laborator 15%
- Lucrare control 15%
- Prezentare si sustinere proiect 20%
- Examen 50%
- Bonus prezenta – 10%

Conditii pentru promovare:

- Nota de la laborator trebuie sa fie minim 5 (prezenta obligatorie la toate sedintele de laborator);
- Nota proiect minim 5 (conditie obligatorie pentru intrare in examen);
- Nota de la examen trebuie sa fie minim 5;
- Nota de minim 5 la Lucrarea de control.

In cazul reluarii disciplinei intr-un alt an universitar, activitatile nepromovate trebuie parcurse din nou.

Cerinte proiect

1). Realizarea suportului scris (format pdf) care sa contina capitolele:

Proiectarea Hardware si Proiectarea Software

- In cap. Proiectarea Hardware trebuie sa fie inclusa argumentarea pentru logica de conectare a circuitelor de memorie si I/O, respectiv a perifericelor;
- Este **obligatorie** schema de conectare finala a tuturor componentelor (eventual fara circuitele de alimentare) cu precizarea tuturor pinilor circuitelor folosite si a modului de conectare a acestora; va fi prezentat modul de programare a tablei prin folosirea tastelor (manualul de utilizare);
- In cap. Proiectarea Software trebuie sa fie inclusa logica de constructie a programului software cu toate componentele sale (prin descrieri explicative si organigrame), respectiv listingul programului;
- Este **obligatorie** lista de variabile folosite sub forma unui tabel de forma:

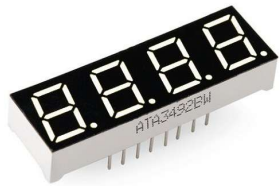
Nume variabila	Semnificatie	Valoarea de initializare	Mod de reprezentare	Adresa de stocare in memorie RAM

- Folosirea unui simulator pentru programul (sau portiuni din el) realizat este un bonus.

2). Predarea proiectului se va face pana in ziua de **23 mai 2026 ora 23:00** pe platforma moddle.

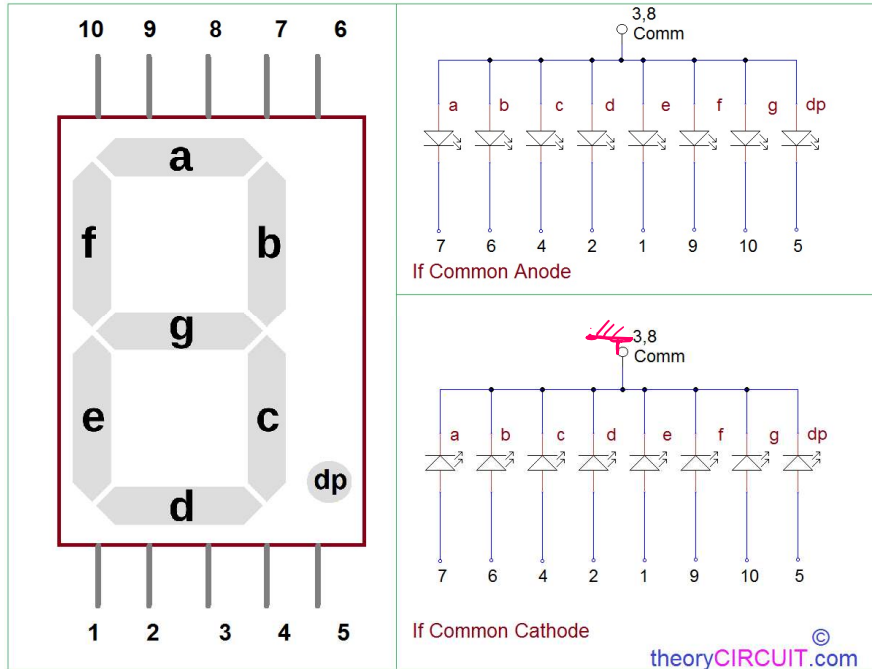
3). Sustinerea proiectului de catre realizator (activitate obligatorie) printr-o sesiune de intrebari legate de solutiile folosite (in ultima saptamana din semestru 25 – 29 mai 2026 la o data stablita de comun acord).

Celule de afisaj LED cu 7 segmente



Celule de afisaj LED cu 7 segmente

7 Segment Display Pinout

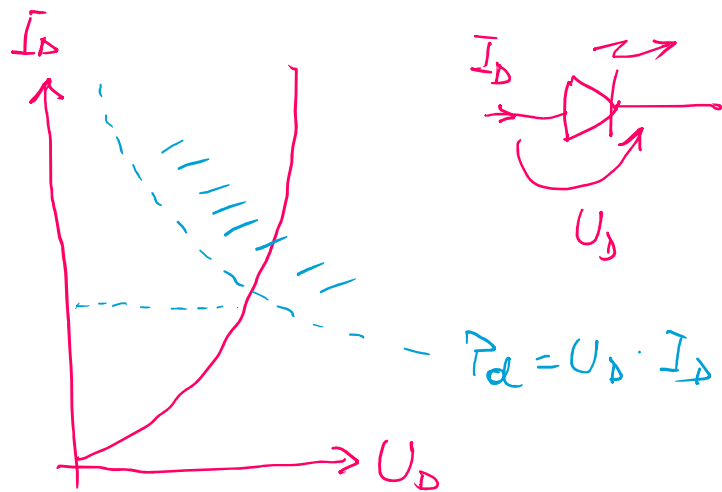
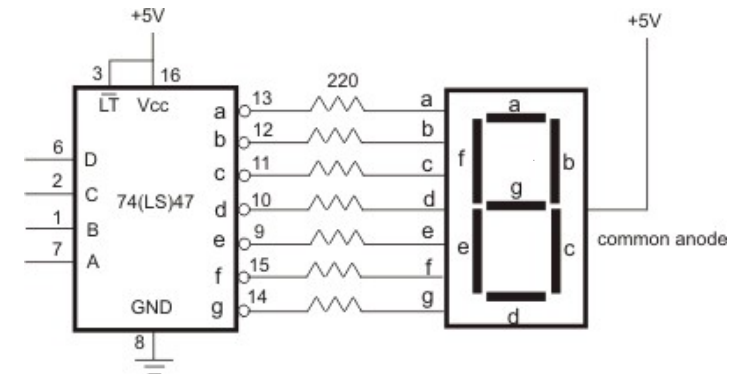
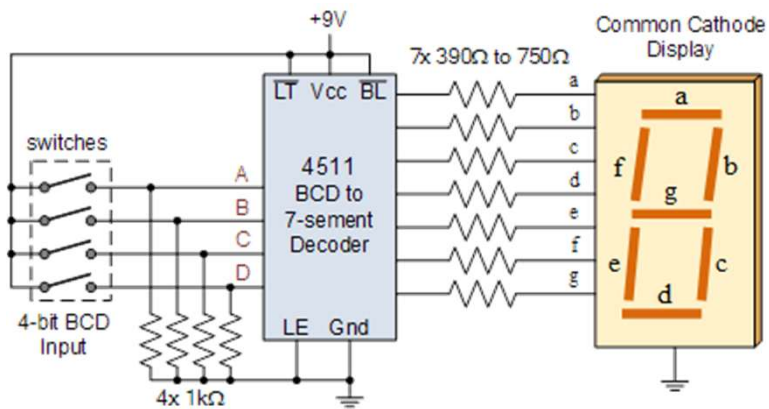


CATHODE COMMON

segment outputs							display
a	b	c	d	e	f	g	
1	1	1	1	1	1	0	0
0	1	1	0	0	0	0	1
1	1	0	1	1	0	1	2
1	1	1	1	0	0	1	3
0	1	1	0	0	1	1	4
1	0	1	1	0	1	1	5
0	0	1	1	1	1	1	6
1	1	1	0	0	0	0	7
1	1	1	1	1	1	1	8
1	1	1	0	0	1	1	9

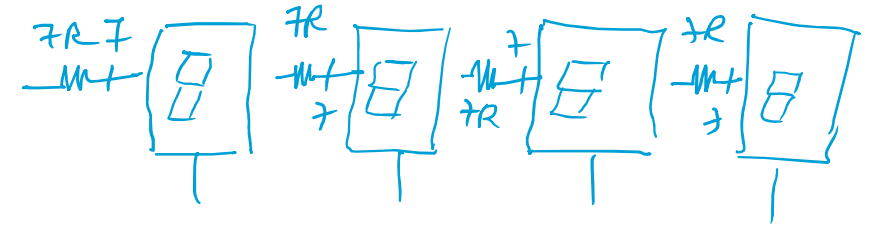
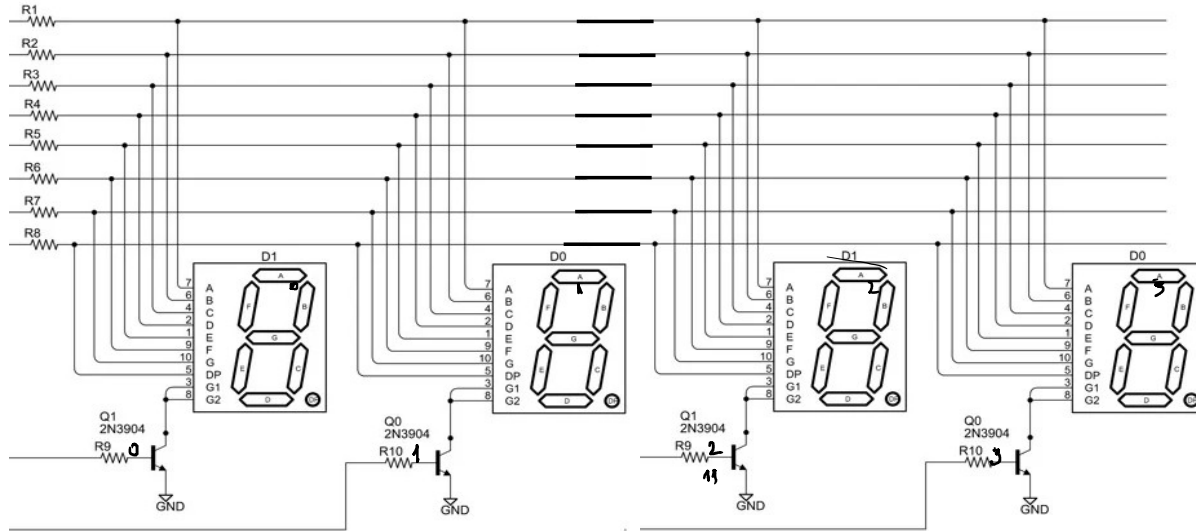
7-Segment Display

Celule de afisaj LED cu 7 segmente



comanda clasic pentru
celula LED 7 segmente -

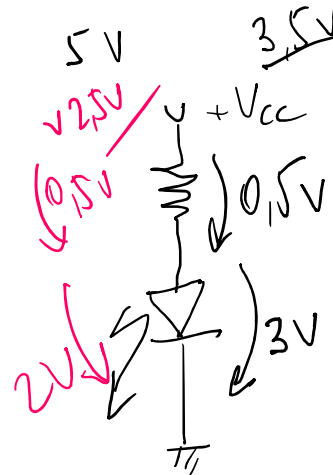
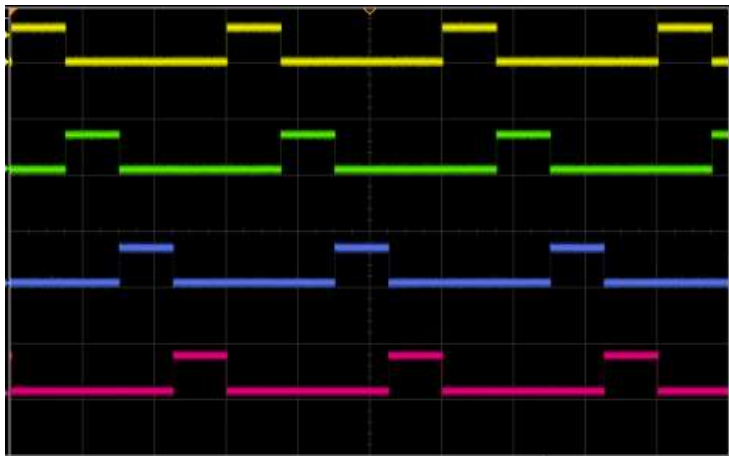
Proiectarea afisajului cu celule LED 7 segmente comandat prin multiplexare



$7R \times 4 = 28R$ Rezistente
 $7 \times 4 = 28$ fire comanda
 4 - catod -

(comanda clasic)

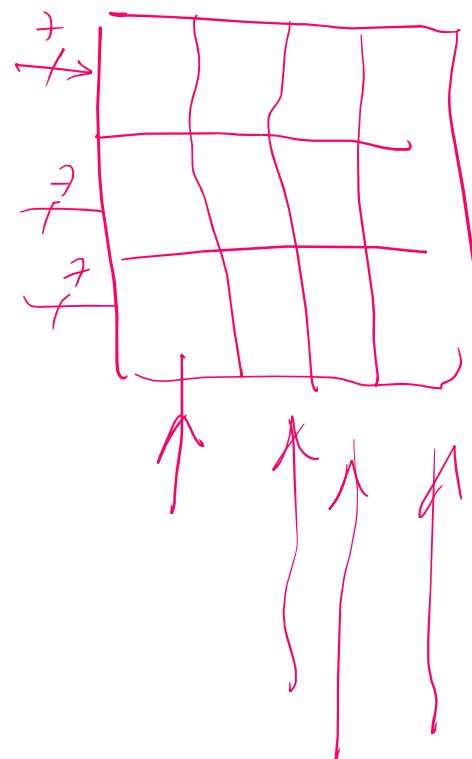
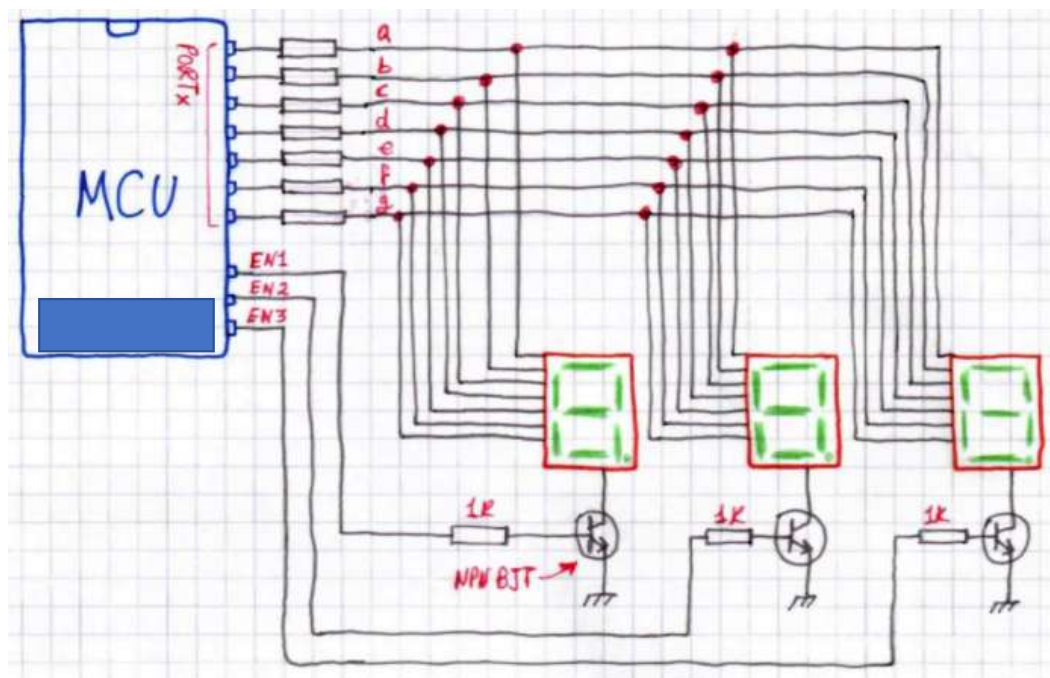
$t = ?$ $f > 50 \text{ Hz}$ $T < 20 \text{ ms}$

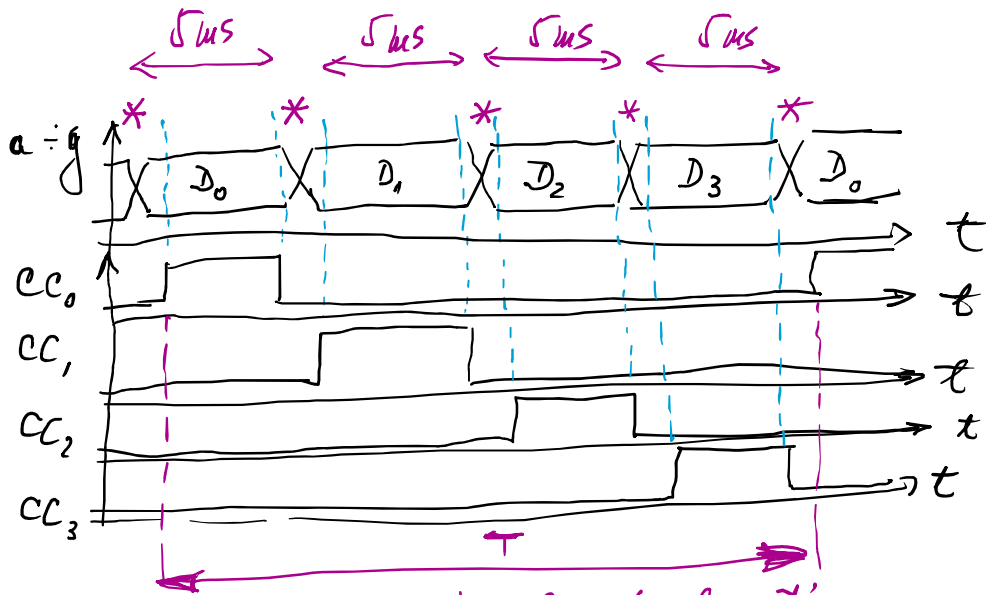


Comanda prin multiplexare
 Resurse

- $7R$ - rezistente
 - $7 + 4 = 11$ fire comanda

Proiectarea afisajului cu celule LED 7 segmente comandat prin multiplexare



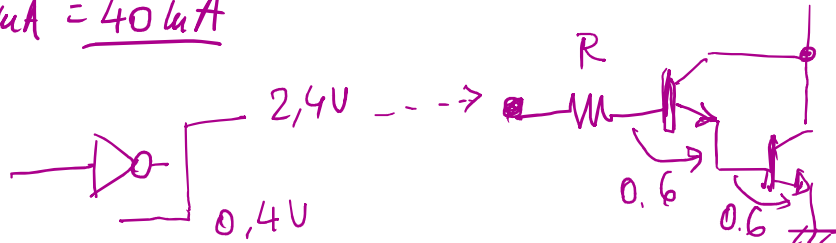


$T \leq 20 \text{ ms}$ ($f \geq 50 \text{ Hz}$) $n = 4$ digits
 $T/n = \frac{20 \text{ ms}}{4} = 5 \text{ ms} \Rightarrow$ necesitate folosire timer

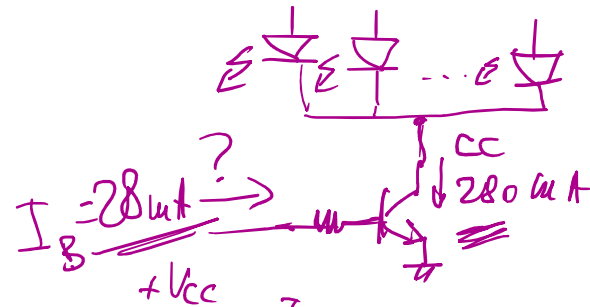
$\Rightarrow$ semnalizari la 5 ms prin cereri de intrerupere catre CPU

$$I_{\text{peak}} \geq n \cdot I_D = 4 \cdot 10 \text{ mA} = 40 \text{ mA}$$

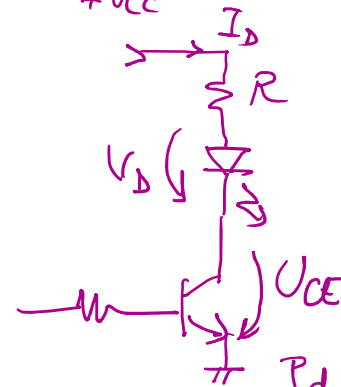
$\Rightarrow$ necesitate circuite amplificari curent



Circuitul de amplificari cu curent C_{T2} (cotelod comun) —



medie putere BD 141, 143,



$$V_{CC} = I_D \cdot R + V_D + U_{CE}$$

$$5 = 40 \cdot R + 2 + U_{CE \text{ SAT}}$$

$$R = \frac{5 - 2 - 1}{40} = \frac{2}{40} = 0,05 \text{ k}$$

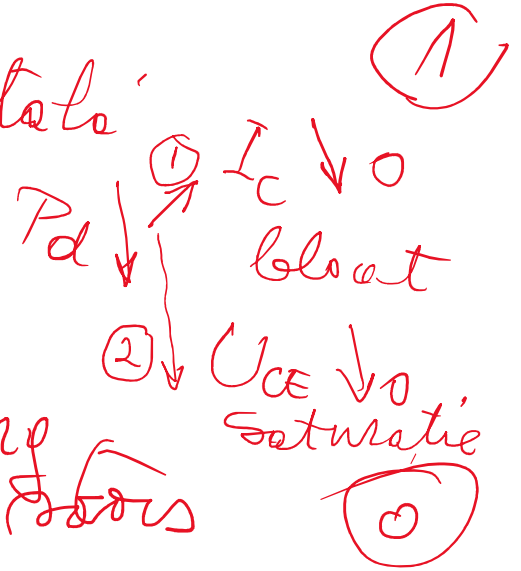
$$\beta_{\text{SAT}} = 10!$$

Regim saturat $U_{CE} \rightarrow 0$
 Regim blocat $I_C \rightarrow 0$

Operare cu tranzistori în logica digitală

1) Putere disipată

$$P_{d\text{tranz}} = U_{CE} \cdot I_C$$



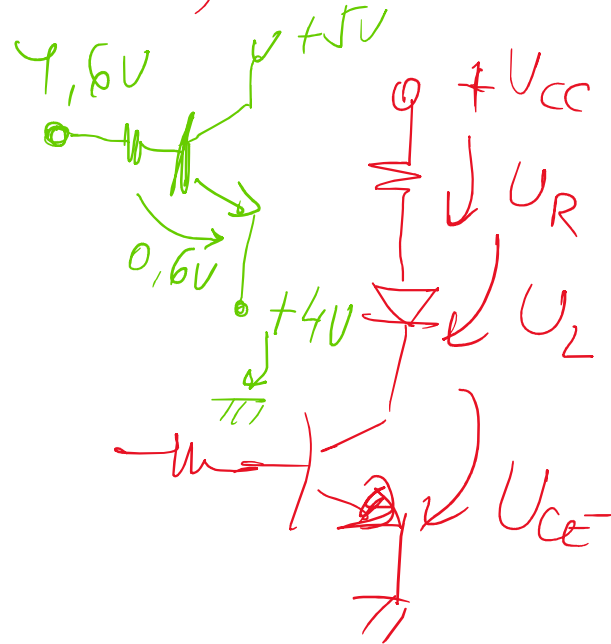
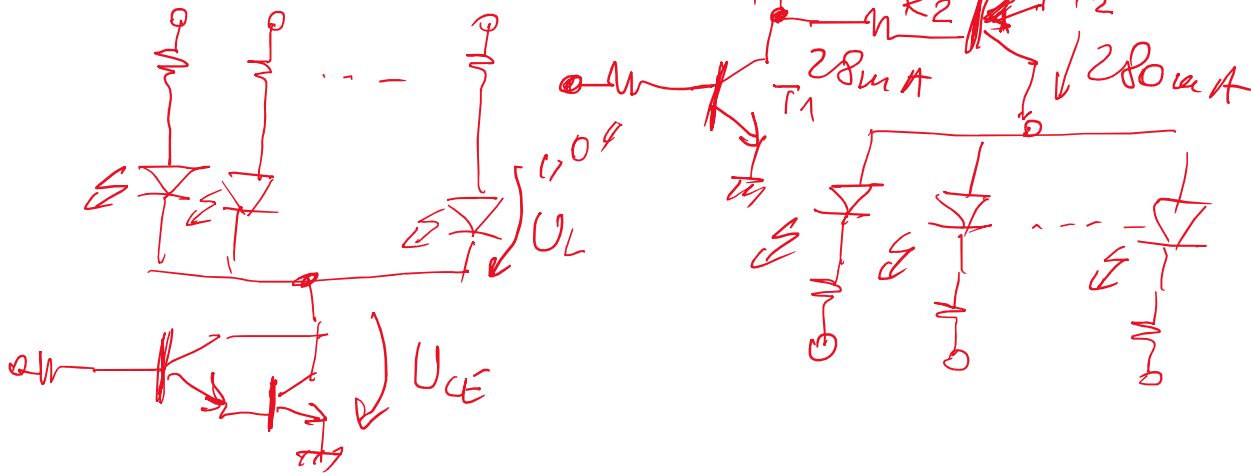
Switching transistors

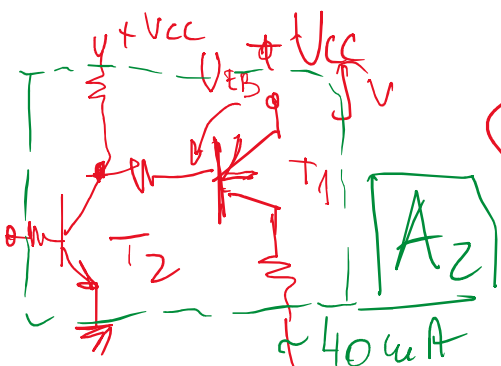
$$P = U \cdot I = I \cdot R \cdot I = I^2 \cdot R$$



Catod comun

Anod comun

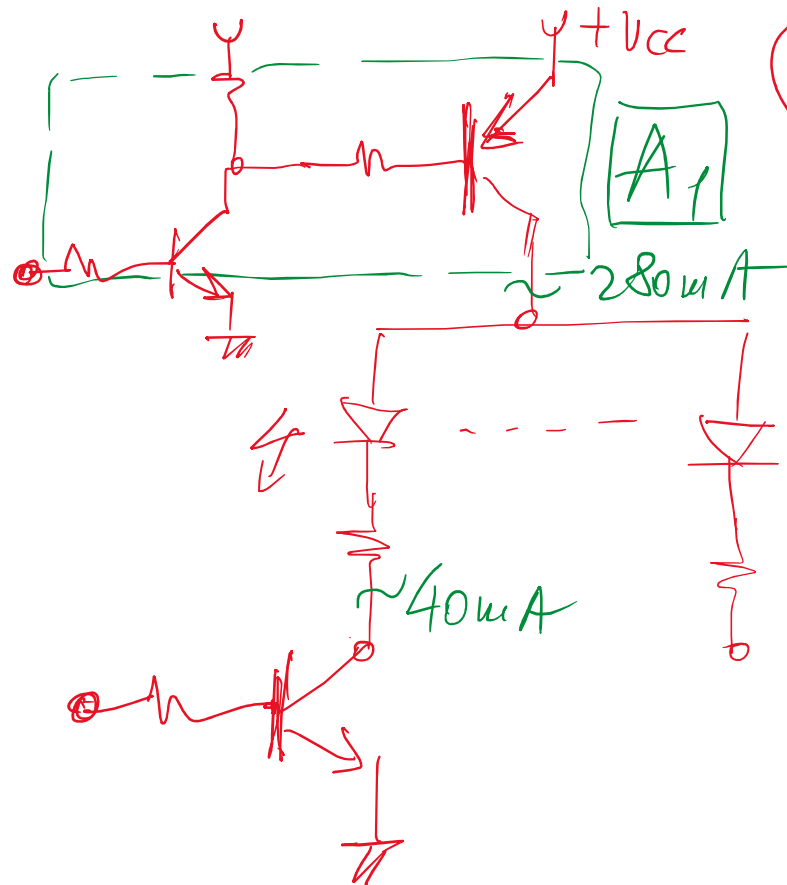
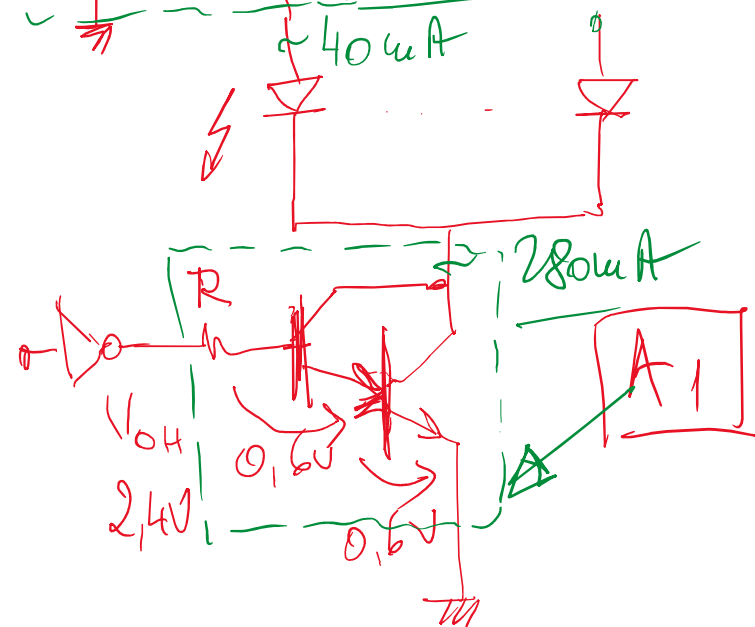




CC

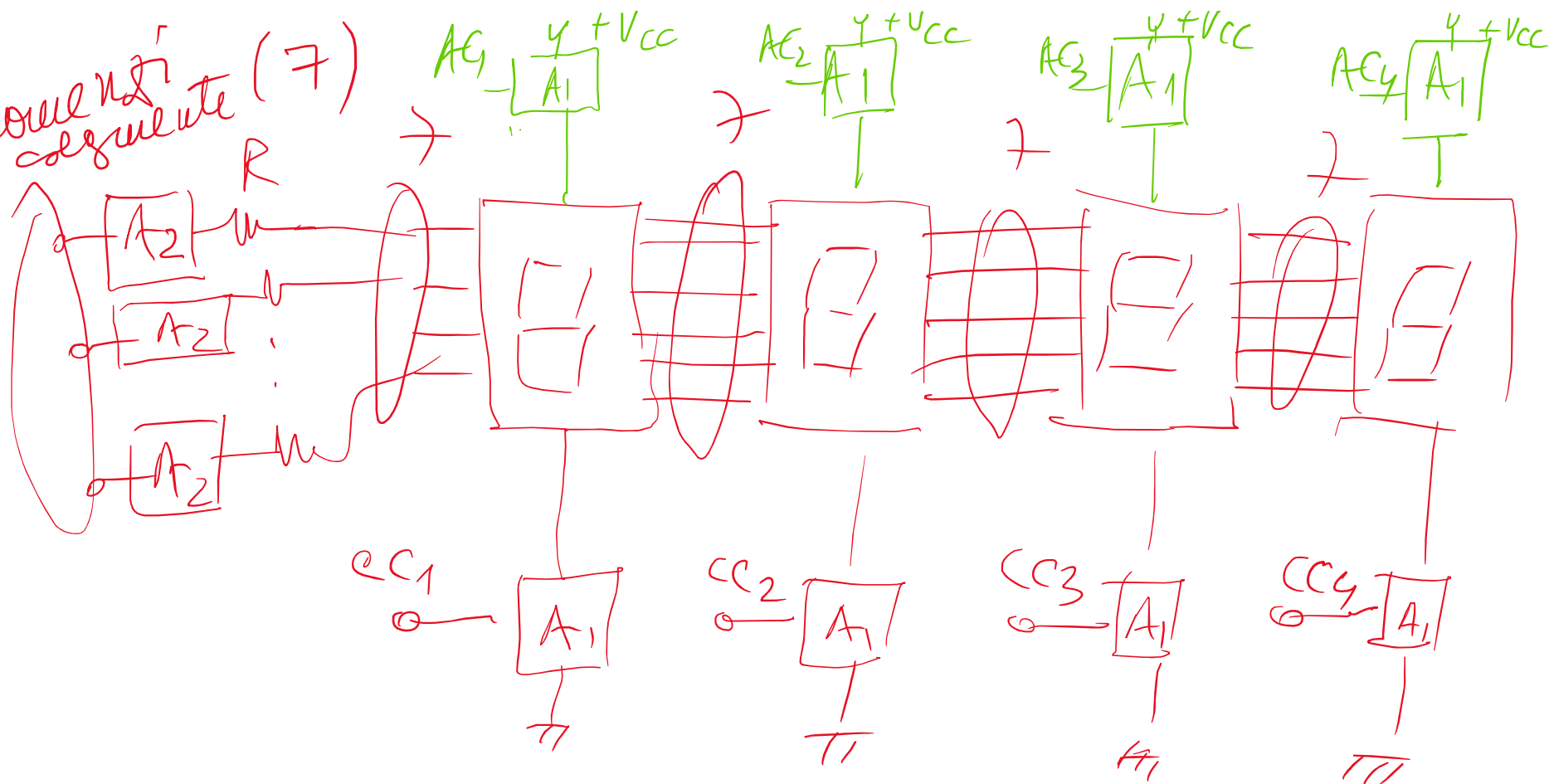
$$V_{OL} \leq 0,4V$$

$$V_{OH} \geq 2,4V$$



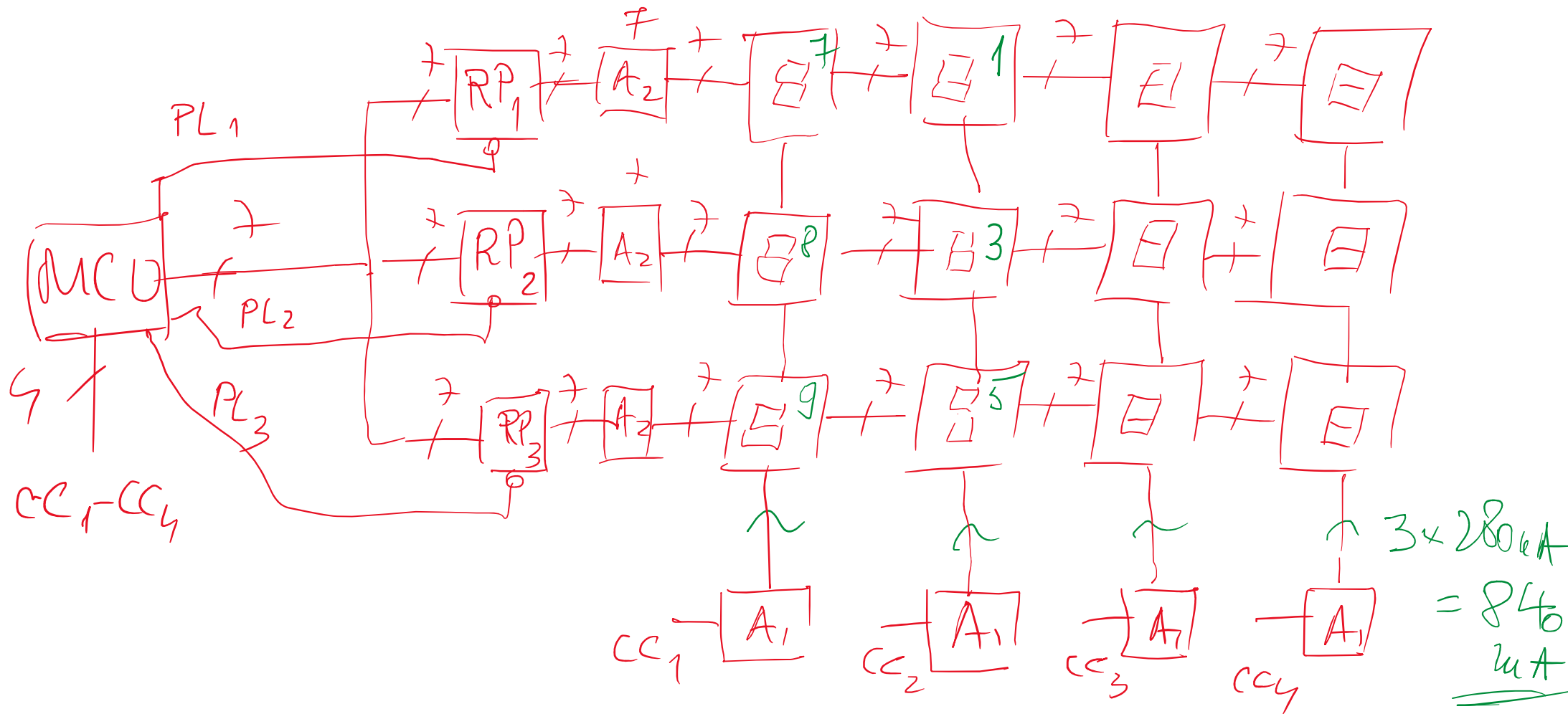
AC

could not
converge (7)



AE

PL-parallel load

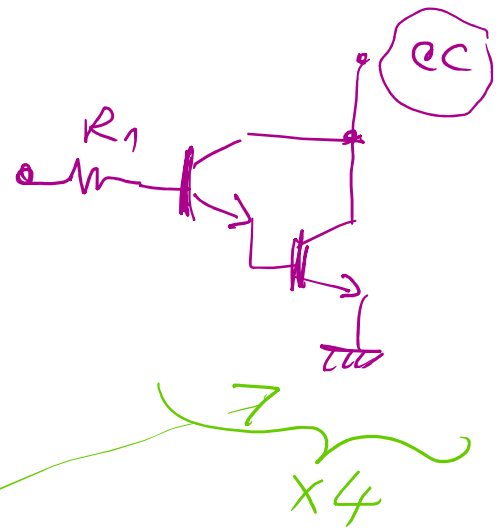
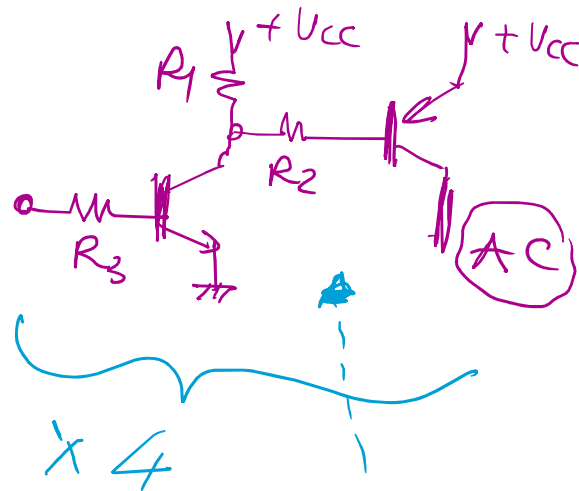
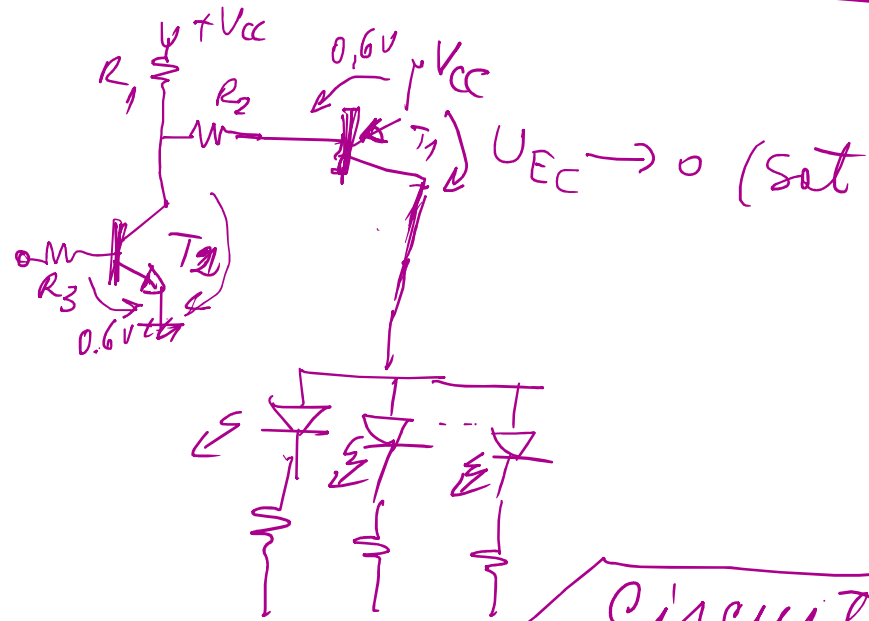


SURSE

- 1) Căutare celule LED 7seg (www.farnell.com)
- 3) Căutare tranzistoare (—||—) pentru A_1 și A_2
- 2) Scheme principale de multiplexare pentru matricea de celule 7seg din tablă (calcul curent)
- 4) Proiectarea A_1 și A_2 (valori de rezistențe + alegere tranzistor)

ANOD COMUN

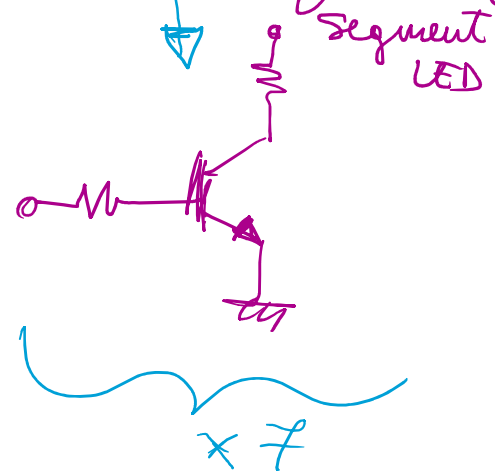
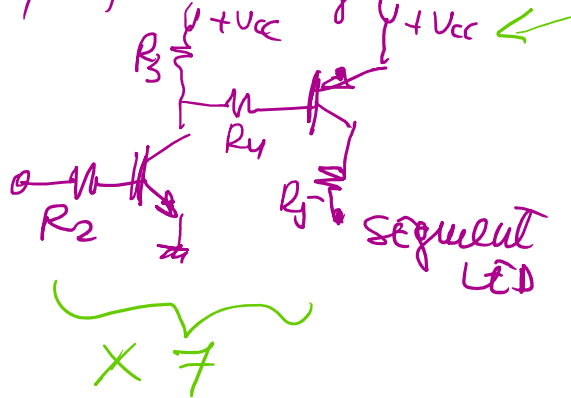
CA2

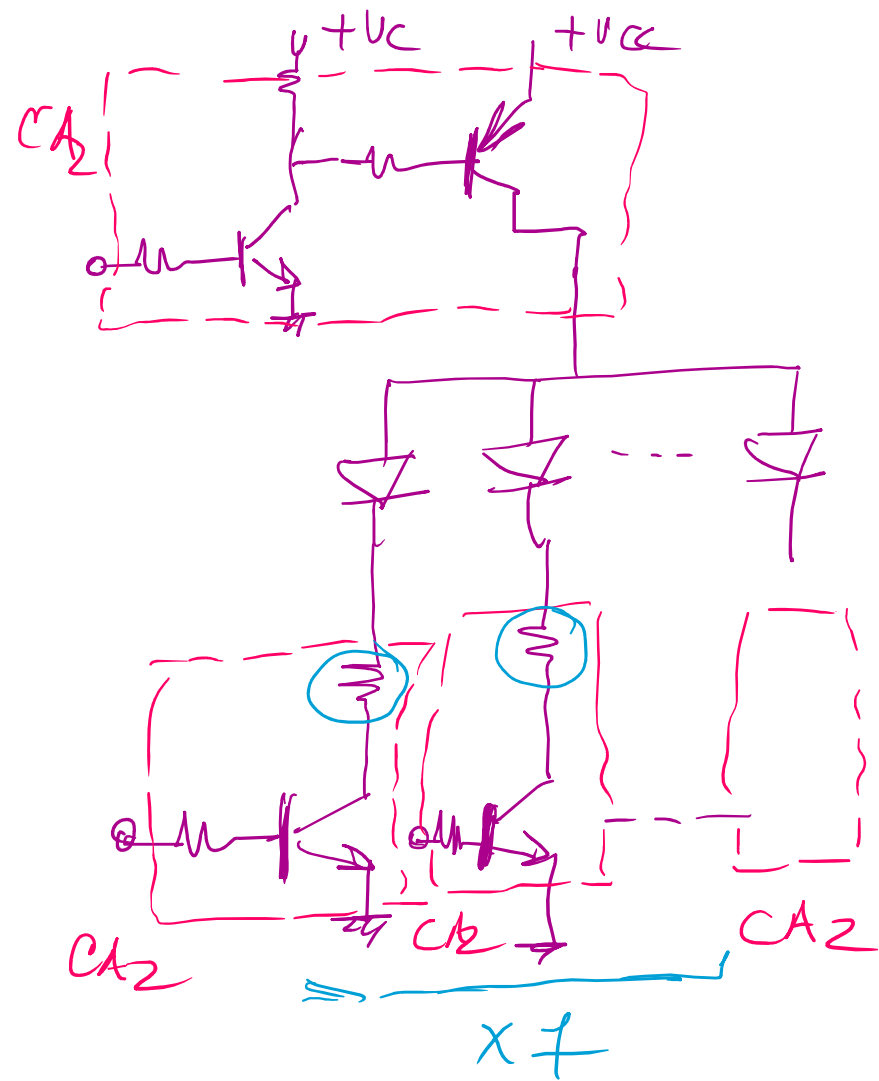
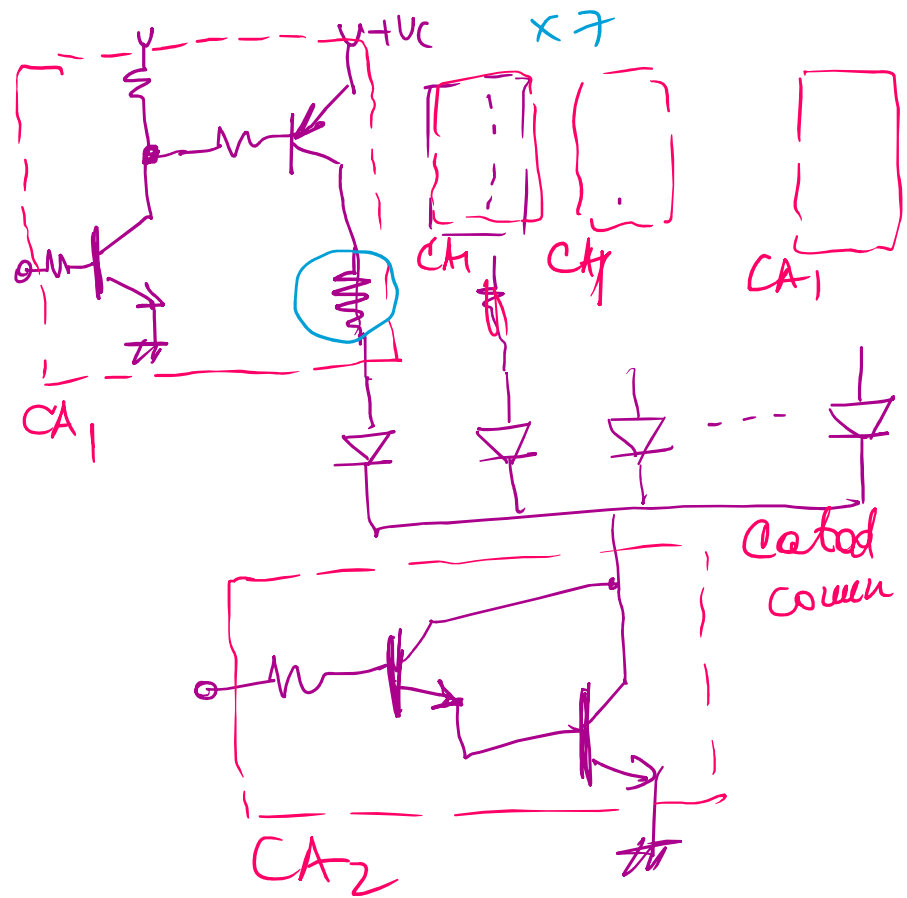


Circuit amplifcare curent pe segmente (CA1)

CATOD COMUN

pe fiecare segment



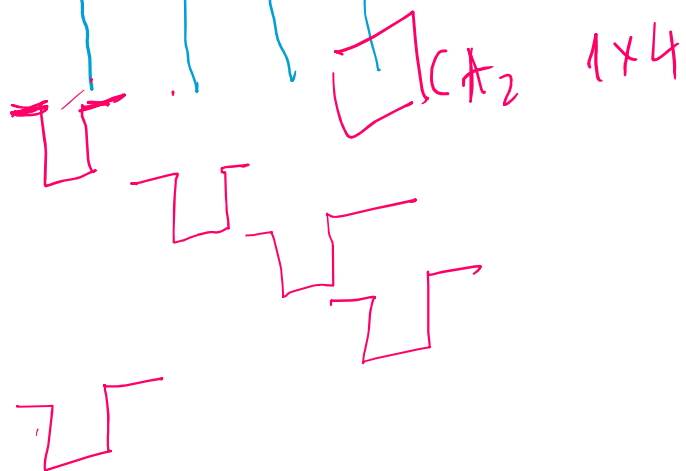
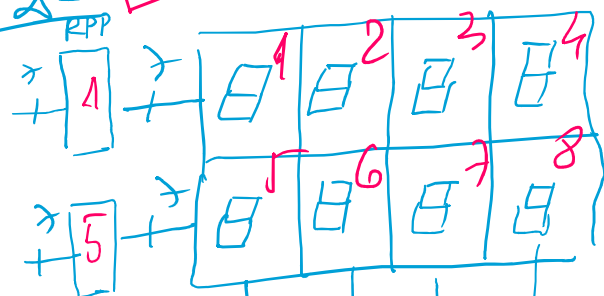


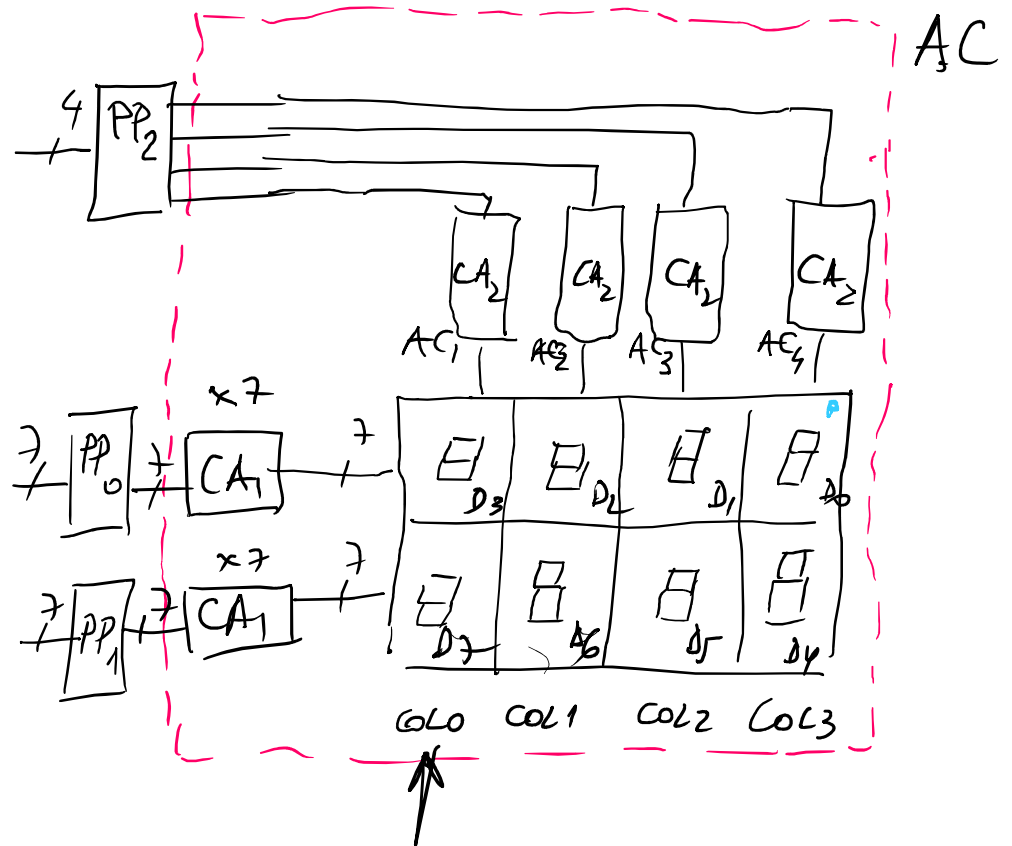
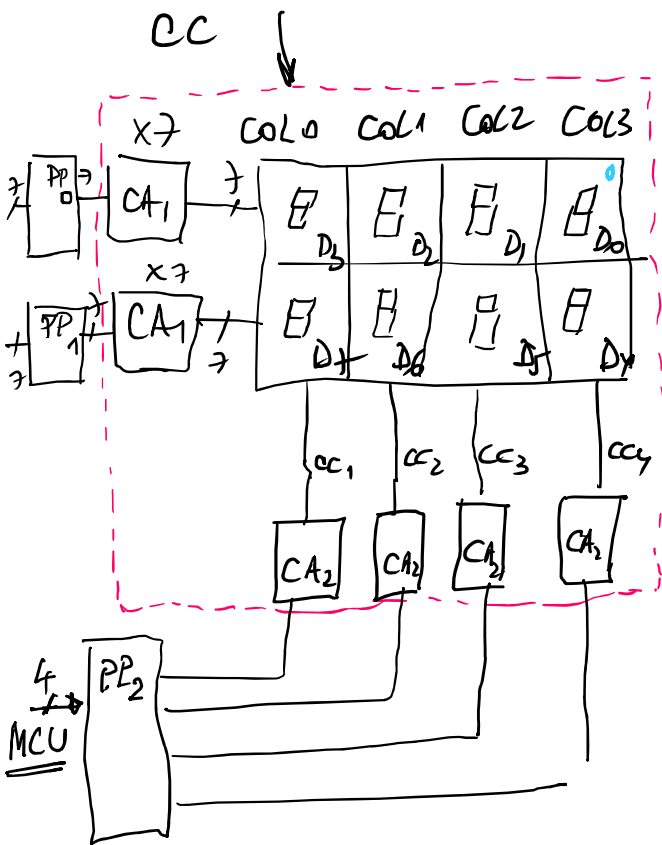
Disponere celule afroaj

caz 1



caz 2 $\square CA_1 \rightarrow 2 \times 4$





END 4 mar 2026